Integration

Chapter Overview

Recover a function from its derivative

Integrate powers and sums. Use a point on a curve to determine the constant of integration.

Calculate areas

Find areas between curves, lines and the x-axis. Split the calculation when a curve crosses the axis.

Evaluate definite integrals

Use upper and lower limits to calculate a numerical result.

Connect differentiation and integration

Understand why integration recovers an area from its rate of change.

Integrating x^n terms 3

Integration reverses differentiation.

$5x^3 \xrightarrow{\text{differentiate}} 15x^2$

The power rule

$$\int ax^n \, dx = \frac{ax^{n+1}}{n+1} + c \qquad (n \neq -1)$$

Increase the power by one, then divide by the new power.

If $dy/dx = 3x^2$, find y.

Answer

The derivative of every constant is zero. The constant c represents the whole family of possible functions.

Examples 4

Find y for each derivative. Include the constant of integration.

1.

$$\frac{dy}{dx} = 4x^3$$

1

2.

$$\frac{dy}{dx} = x^5$$

2

3.

$$\frac{dy}{dx} = 3x^{1/2}$$

3

Check by differentiating your answer. Fractional and negative powers obey the same rule, except for power -1.

More Fractional Examples 5

Find y for each derivative. Include the constant of integration.

1. $\frac{dy}{dx} = \frac{4}{\sqrt{x}}$

1

2. $\frac{dy}{dx} = 5x^{-2}$

2

3. $\frac{dy}{dx} = 4x^{2/3}$

3

4. $\frac{dy}{dx} = 10x^{-2/7}$

4

Check by differentiating your answer. Fractional and negative powers obey the same rule, except for power -1.

Test Your Understanding 6

Find f(x) from each derivative. For 3 and 4, reveal the rewrite before the integral.

1. $f\prime(x) = 2x + 7$

1: integral

2. $f\prime(x) = x^2 - 1$

2: integral

3. $f\prime(x) = \frac{2}{x^7}$

3: rewrite

3: integral

4. $f\prime(x) = \sqrt[3]{x}$

4: rewrite

4: integral

5. $f\prime(x) = 33x^{5/6}$

5: integral

The power rule excludes $x^{-1} = 1/x$: adding one would give a zero denominator. Its integral is studied in Year 2.

Exercise 13A – Y1 pages 289-290 7

4. Find f(x) when $f\prime(x) = (2x + 3)^2$.

5. Find f(x) when $f\prime(x) = 3x^{-2} + 6\sqrt{x} + x - 4$.

4: expand

5: rewrite

4: integrate

5: integrate

Exercise 13A – Y1 pages 290 8

Challenge: integrate $\frac{(2\sqrt{x} - x^2)(3 + x)}{x^5}$.

Expand and divide each term by x^5

Integrate term by term

Integration notation 9

Notation for integration

$$\frac{d}{dx}(5x^2) = 10x \qquad \Longleftrightarrow \qquad \int 10x\,dx = 5x^2 + c$$

The integral sign

$\int$ tells us to integrate. An indefinite integral has no limits and represents a family of antiderivatives.

The variable

The dx tells us to integrate with respect to x. For example, $\int 6t^2\,dt = 2t^3 + c$.

Use brackets when the integrand contains several terms. Remember c for an indefinite integral.

Examples 10

Integrate each expression. In 3, p and q are constants.

1. $\int (x^{-3/2} + 2)\,dx$

1

2. $\int (6t^2 - 1)\,dt$

2

3. $\int (px^3 + q)\,dx$

3

Test Your Understanding 11

Given $y = 2x^5 + 6/\sqrt{x}$, where x > 0, find $\int y\,dx$.

Write the root as a power

Integrate

Exercise 13B – Y1 pages 291-293 12

Integrate each expression.

10. $$\int \left(\frac{6}{x^2} + 4\sqrt{x} - 3x + 2 \right) dx$$

10

11. $$\int \left(8x^3 + 6x - \frac{3}{\sqrt{x}} \right) dx$$

11

Exercise 13B – Y1 pages 291-293

12. (a) Write $(2+5\sqrt{x})^2$ as $4+k\sqrt{x}+25x$.

12a

(b) Hence integrate the expression.

12b

13. Given $y = 3x^5 - 4/\sqrt{x}$, find $\int y\,dx$.

13: rewrite

13: integrate

Exercise 13B – Y1 pages 291-293

14. Find p and q if $\int (\frac{p}{2x^2} + pq)\,dx = \frac{2}{x} + 10x + c$.

14

15. (a) Expand $(2-x)^{10}$, retaining terms up to x^2.

15a

(b) Integrate these terms to obtain an approximation near $x = 0$.

15b

Finding constant of integration 15

Finding the constant of integration

All functions $y = x^3 + c$ have derivative $3x^2$. A known point selects one member of this family.

Find f(x) if $f\prime(x) = 3x^2$ and the curve passes through (1,3).

Integrate

Substitute the point

State the function

Test Your Understanding 16

The curve has derivative $f\prime(x) = \frac{3}{8}x^2 - 10x^{-1/2} + 1$ for x > 0 and passes through (4,25).

(a) Find f(x).

a

(b) Find the normal at (4,25).

b

4. A curve passes through $(-1,0)$ and has $f\prime(x) = 9x^2 + 4x - 3$. Find f(x).

4

5. Find y if $dy/dx = 3x^{-1/2} - 2x\sqrt{x}$ and y = 10 when x = 4.

5

6. A curve has $dy/dx = (6x + 5x^{3/2})/\sqrt{x}$ and y = 100 at x = 9.

(a) Write the derivative as $6x^p + 5x^q$.

a

(b) Find the equation of the curve.

b

Definite Integration 19

Definite integration

A definite integral has lower and upper limits. Evaluate an antiderivative at the upper limit, then subtract its value at the lower limit.

$$\int_1^5 4x^3\,dx$$

Example

Connection with area

For a nonnegative function, the definite integral gives the area under the graph between the limits. For a speed–time graph, this area represents distance travelled.

Another Example 20

Evaluate $\int_{-3}^{3}(x^2+1)\,dx$.

Integrate and substitute both limits

Answer

Why is there no c?

$$[F(x)+c]_a^b = F(b)+c-F(a)-c$$

The constants cancel. Use brackets around the whole lower-limit value, especially when it is negative.

Problem Solving 21

Find P if $\int_1^5 (2Px + 7)\,dx = 4P^2$.

Exercise 13D – Y1 page 297 22

4. Find A if $\int_1^4 (6\sqrt{x} - A)\,dx = A^2$.

4

4

5. Evaluate $\int_1^9 (2x - 3\sqrt{x})\,dx$.

5

6. Express $\int_4^{12} \frac{2}{\sqrt{x}}\,dx$ in the form a + b$\sqrt{3}$.

6

7. Find k if $\int_1^k \frac{1}{\sqrt{x}}\,dx = 3$.

7

7

EXTENSION 25

1. [MAT 2009 1A] Find the smallest value of $I(a) = \int_0^1 (x^2 - a)^2 \, dx$ as a varies.

1

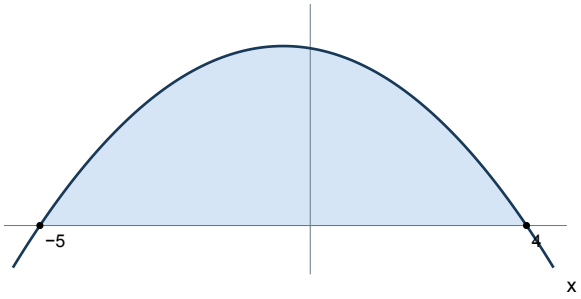
2. [MAT 2015 1D] Let $f(x) = \int_0^1 (xt)^2 \, dt$ and $g(x) = \int_0^x t^2 \, dt$. For A > 0, which is greater: g(f(A)) or f(g(A))?

2

Areas under curves 26

For a curve above the x-axis, its area between x = a and x = b is

$$\int_a^b f(x) \, dx$$



Find the finite area bounded by $y = 20 - x - x^2$ and the x-axis.

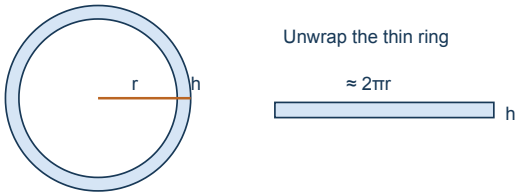
Limits

Area

Just for your interest... 27

Why does integration give area?

A circle has area $A = \pi r^2$, so $dA/dr = 2\pi r$, its circumference.



Increase the radius by a small amount h.

$\pi(r+h)^2 - \pi r^2 = 2\pi rh + \pi h^2$

The thin ring is approximately a strip of length 2πr and width h.

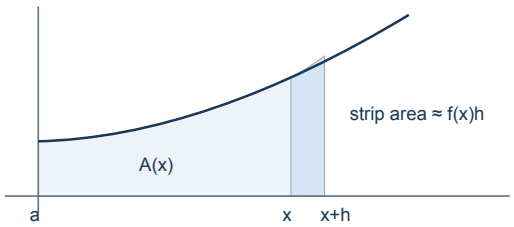
Divide by h and let h approach zero.

$$\frac{\Delta A}{h} = 2\pi r + \pi h \longrightarrow 2\pi r$$

Adding these thin rings is integration: it recovers the area from its rate of change.

Why integration gives area 28

For a sphere, $V = \frac{4}{3}\pi r^3$, so $dV/dr = 4\pi r^2$, its surface area.



For a graph, let A(x) be the accumulated area from a fixed starting point to x.

A thin strip of width h has approximate area $f(x)h$.

$$\frac{A(x+h) - A(x)}{h} \longrightarrow f(x)$$

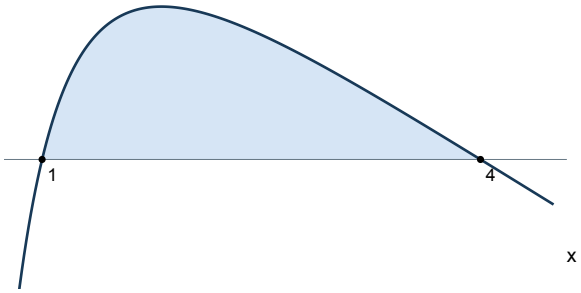
Therefore differentiation of accumulated area gives the original function.

$A\prime(x) = f(x)$

$$\int_a^b f(x)\,dx = A(b) - A(a)$$

Test Your Understanding 29

Find the area under $y = 27 - 2x - 9\sqrt{x} - 16/x^2$ between x = 1 and x = 4.

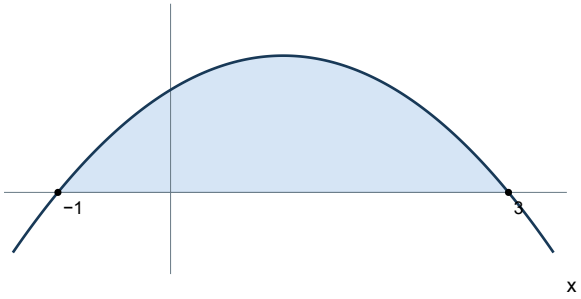


Set up

Evaluate

Exercise 13E – Y1 pages 299-300 30

8. The curve $y = -x^2 + 2x + 3$ bounds a finite region with the x-axis. Find its area.

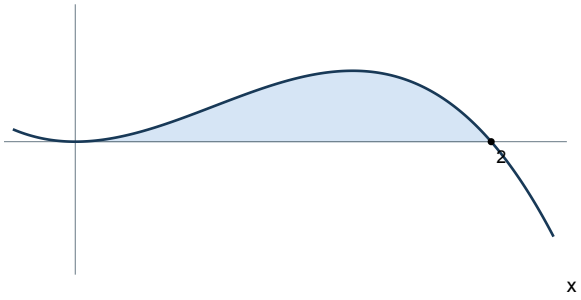


Limits

Area

Exercise 13E – Y1 pages 299-300

9. Find the finite area bounded by $y = x^2(2 - x)$ and the x-axis.



Limits

Area

EXTENSION

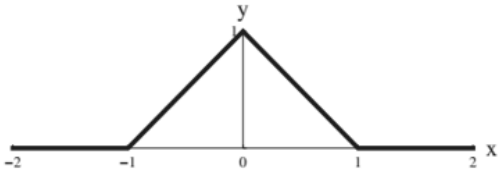
1. [MAT 2007] Given

$$3\int_0^1 f(x)\,dx + 2\int_1^2 f(x)\,dx = 7$$

$$\int_0^2 f(x)\,dx + \int_1^2 f(x)\,dx = 1$$

find $\int_0^2 f(x)\,dx$.

2. [MAT 2011] Using the graph of f, evaluate $\int_{-1}^{1} f(x^2 - 1)\,dx$.



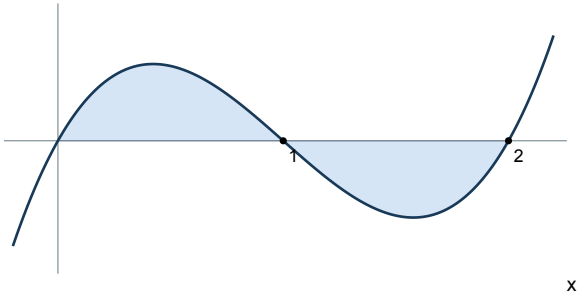
1

2

‘Negative Areas’ 33

Definite integrals give signed area.

Area above the x-axis contributes positively. Area below it contributes negatively.



Consider $f(x) = x(x-1)(x-2)$.

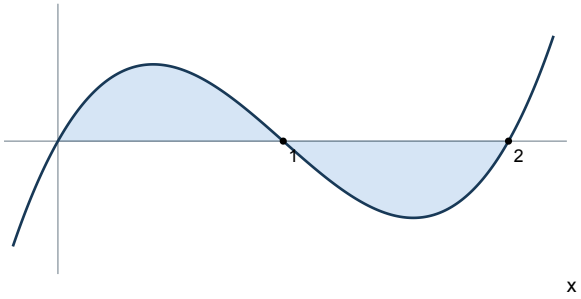
Does zero mean there is no area?

Integral

Meaning

Example 34

Find the total area between $y = x(x-1)(x-2)$ and the x-axis from x = 0 to x = 2.



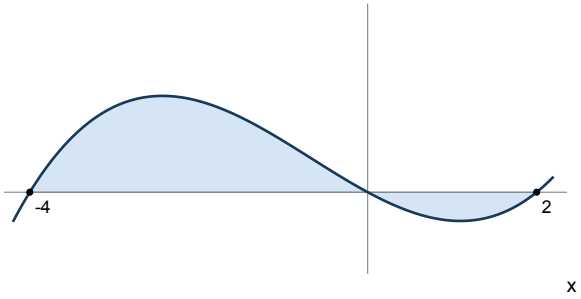
Above the axis

Below the axis

Total

Test Your Understanding 35

The curve is $y = x(x+4)(x-2)$. (a) Find its x-intercepts. (b) Find the total area of the finite regions bounded by the curve and the x-axis.



a

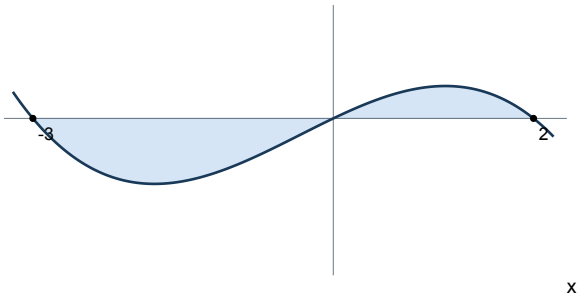
Split at zero

Integrate

b

Exercise 13F – Y1 pages 301-302 36

2. The curve is $y = x(x+3)(2-x)$. (a) Find its x-intercepts. (b) Find the total area of the finite regions bounded by the curve and the x-axis.



a

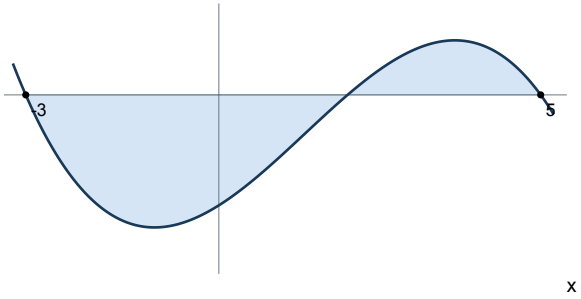
Split at zero

Integrate

b

Exercise 13F – Y1 pages 301-302

3. Let $f(x) = -x^3 + 4x^2 + 11x - 30$. Show that x + 3 is a factor, factorise fully, find all roots, and calculate the total area between the curve and the x-axis.



Factorise

Area

EXTENSION

[MAT 2010] For a > 0, where is $I(a) = \int_0^a (4 - 2^{x^2})\,dx$ stationary?

Answer

[STEP 2014 Q3] Let b > a ≥ 0 and

$$\int_a^b x^2\,dx = \left(\int_a^b x\,dx\right)^2$$

(i) Find b when a = 0. (ii) When a = 1, show that b satisfies a cubic with exactly one real root, between 2 and 3. (iii) Put p = b + a and q = b − a. Show that $3p^2 + q^2 = 3p^2q$ and deduce $1 < q \leq 4/3$.

Worked solution on the next slide.

Guidance for Extension Problem 2 — 39

(i) $a = 0$

i

ii: sketch

(ii) $a = 1$

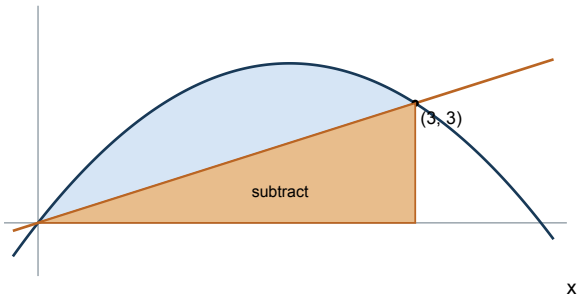
ii

(iii) $p = b + a$, $q = b - a$

iii

Areas between curves and lines — 40

Find the area between $y = x(4 - x)$ and $y = x$.



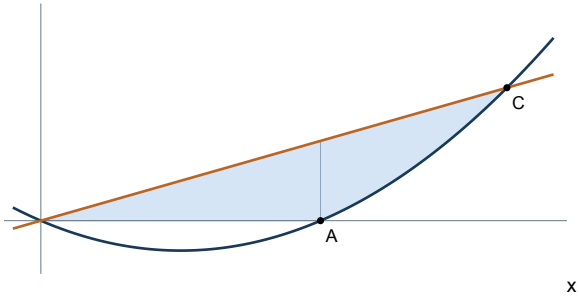
1. Intersections

2. Area under curve

3. Subtract triangle

A Harder One 41

The curve is $y = x(x-3)$ and the line is $y = 2x$. Find the region above the x-axis, below the line and to the left of the positive branch of the curve.



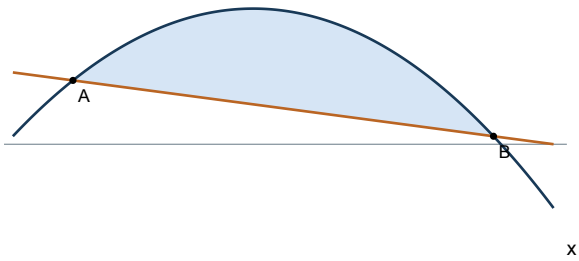
1. Points

2. Strategy

3. Area

Test Your Understanding 42

The curve $y = 10x - x^2 - 8$ and line $y = 10 - x$ enclose a region. (a) Find their intersections. (b) Find the area.



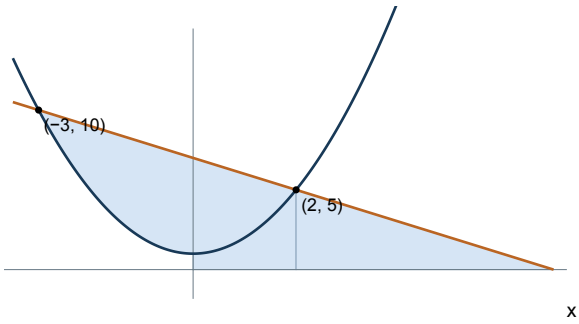
a

b

General rule:

$$\text{Area} = \int_a^b (\text{upper function} - \text{lower function})\, dx$$

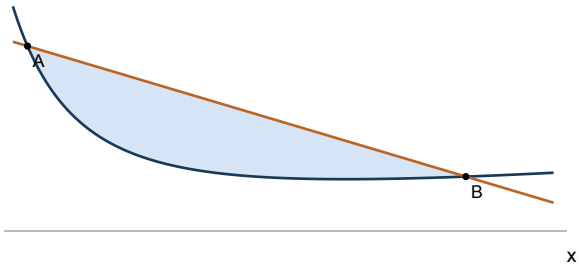
6. The curve $y = x^2 + 1$ and line $y = 7 - x$ enclose R_1. R_2 is the region in the first quadrant below both graphs. Find both areas.



R_1

R_2

8. Points A and B on $y = 2/x^2 + x$ have x-coordinates $1/2$ and 2. Find the area between chord AB and the curve, to 3 significant figures.

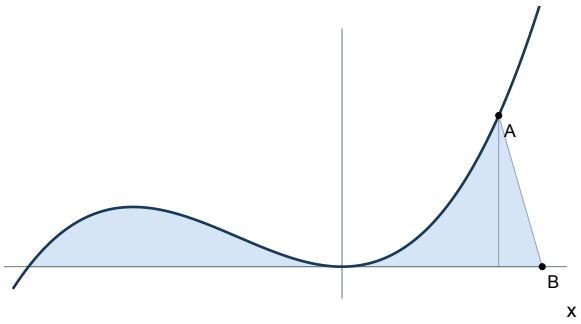


Line

Area

Exercise 13G – Y1 pages 304-306

10. The curve is $y = x^2(x + 4)$. R_1 lies under the curve between -4 and 0. R_2 lies under the curve from 0 to A $= (2, 24)$, then under the straight segment from A to B $= (b, 0)$. Find b if the areas are equal.



R_1

R_2 and b

EXTENSION

a. [MAT 2005] Find the area between $y = x^2$ and $y = x + 2$.

a

b. [MAT 2016] Let $f(x) = a - x^2$ and $g(x) = x^4 - a$, with $a > 0$. When is the finite area bounded by f and the x-axis greater than the corresponding area for g?

b